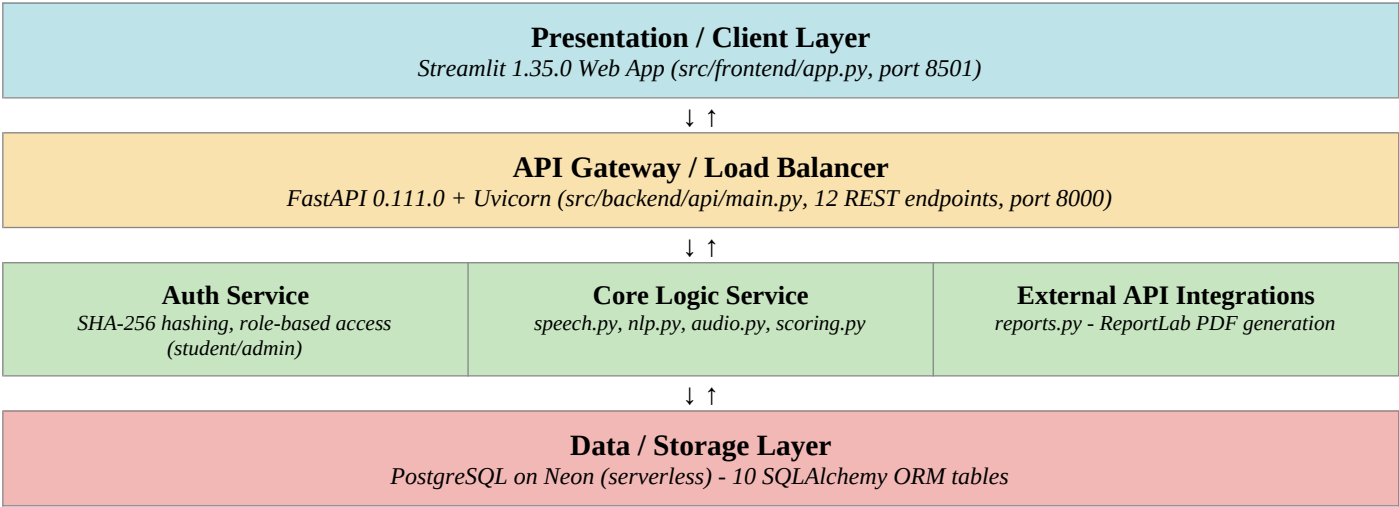


Solution Architecture

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Streamlit UI: authentication forms, concept browser, audio recorder, evaluation scorecard, PDF download, history, admin dashboard	Streamlit 1.35.0, requests, matplotlib, pillow
API Gateway	FastAPI REST coordination layer handling request validation, pipeline orchestration, and error handling across 12 endpoints	FastAPI 0.111.0, Uvicorn 0.29.0, Pydantic
Microservices	AI/ML service layer: Whisper STT, Bi/Cross-Encoder NLP, Librosa DSP, weighted scoring, ReportLab PDF generation	openai-whisper, sentence-transformers, librosa, reportlab
Database	SQLAlchemy ORM with 10 normalised tables and context-managed sessions (get_db())	SQLAlchemy 2.0.30, PostgreSQL (Neon), psycpg2-binary